

Lecture-3
Mathematics 1 (15B11MA111)

CO [C105.1]

Module: Partial differentiation

Topics to be covered

➤ Continuity

- ❖ Definition

- ❖ Examples

- ❖ Assignment

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

➤ Continuity

❖ Definition

If $z = f(x, y)$ be a function of two variables then it is said to be continuous at a point $P(x_0, y_0)$ if

(i) $f(x, y)$ is defined at the point $P(x_0, y_0)$.

(ii) $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y)$ exists, and

(iii) $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y) = f(x_0, y_0)$.

$\delta - \varepsilon$ approach:

A function $f(x, y)$ is said to be continuous at a point $P(x_0, y_0)$ if $|f(x, y) - f(x_0, y_0)| < \varepsilon$, whenever $\sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$.

Removable discontinuity

If $f(x, y)$ be defined at $P(x_0, y_0)$ and $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = L$ exists, but $f(x_0, y_0) \neq L$ then the point $P(x_0, y_0)$ is called a point of removable discontinuity.

It can be continuous by redefining the function as $f(x_0, y_0) = L$

Examples:

Show that the following functions are continuous at the point $(0,0)$

$$i. \quad f(x, y) = \begin{cases} \frac{2x^4 + 3y^4}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

$$ii. \quad f(x, y) = \begin{cases} \frac{2x(x^2 - y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

$$(iii) f(x, y) = \begin{cases} \frac{\sin^{-1}(x+2y)}{\tan^{-1}(2x+4y)}, & (x, y) \neq (0, 0) \\ 1/2, & (x, y) = (0, 0) \end{cases}$$

Solution

(i) Let $x = r\cos\theta, y = r\sin\theta$. Then $r^2 = x^2 + y^2$.

$$|f(x, y) - f(0, 0)| = \left| \frac{2x^4 + 3y^4}{x^2 + y^2} \right|$$

$$= \left| \frac{r^4(2\cos^4\theta + 3\sin^4\theta)}{r^2(\cos^2\theta + \sin^2\theta)} \right| < r^2 [2|\cos^4\theta| + 3|\sin^4\theta|] \\ < 5r^2 < \epsilon$$

or $r = \sqrt{x^2 + y^2} < \sqrt{\frac{\epsilon}{5}}.$

If we choose $\delta < \sqrt{\frac{\epsilon}{5}}$, we find that $|f(x, y) - f(0, 0)| < \epsilon$

whenever $0 < \sqrt{(x)^2 + (y)^2} < \delta$.

Therefore

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y) = f(0, 0) = 0.$$

Hence $f(x, y)$ is continuous at $(0,0)$.

(ii) Let $x = r\cos\theta, y = r\sin\theta$. Then $r^2 = x^2 + y^2$.

$$|f(x, y) - f(0, 0)| = \left| \frac{2x(x^2 - y^2)}{x^2 + y^2} \right|$$

$$= \left| \frac{2r^3(\cos^2\theta - \sin^2\theta)\cos\theta}{r^2(\cos^2\theta + \sin^2\theta)} \right| < |2r \cos 2\theta \cos \theta| \leq 2r < \epsilon$$

or $r = \sqrt{x^2 + y^2} < \varepsilon/2.$

If we choose $\delta < \frac{\varepsilon}{2}$, we find $|f(x, y) - f(0, 0)| < \varepsilon$ whenever

$$0 < \sqrt{(x)^2 + (y)^2} < \delta.$$

Therefore

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y) = f(0, 0) = 0.$$

Hence $f(x, y)$ is continuous at $(0,0)$.

(iii) Hint: put $x + 2y = t$, limit exists and it is equal to $\frac{1}{2}$ (a finite value) and satisfies the required conditions of continuity.

Examples: Show that the following functions are discontinuous at the given points

$$(i) \ f(x, y) = \begin{cases} \frac{x-y}{x+y}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases} \quad \text{at the point } (0,0)$$

$$(ii) \ f(x, y) = \begin{cases} \frac{x^2 - x\sqrt{y}}{x^2 + y}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases} \quad \text{at the point } (0,0)$$

Solution:

(i) For a path $y = mx$, the value of the limit

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x-y}{x+y} = \frac{1-m}{1+m}.$$

which depends on m . Therefore limit does not exist. The given function is not continuous at $(0,0)$.

(ii) For a path $y = m^2 x^2$, the value of the limit is $\frac{1-m}{1+m^2}$

which depends on m . Therefore limit does not exist. The given function is not continuous at $(0,0)$.

Assignment

Discuss the continuity of the following function at the point (0,0)

$$(i) \ f(x, y) = \begin{cases} \frac{x^2+y^2}{xy}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

$$(ii) \ f(x, y) = \begin{cases} \frac{xy(x-y)}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Thank you